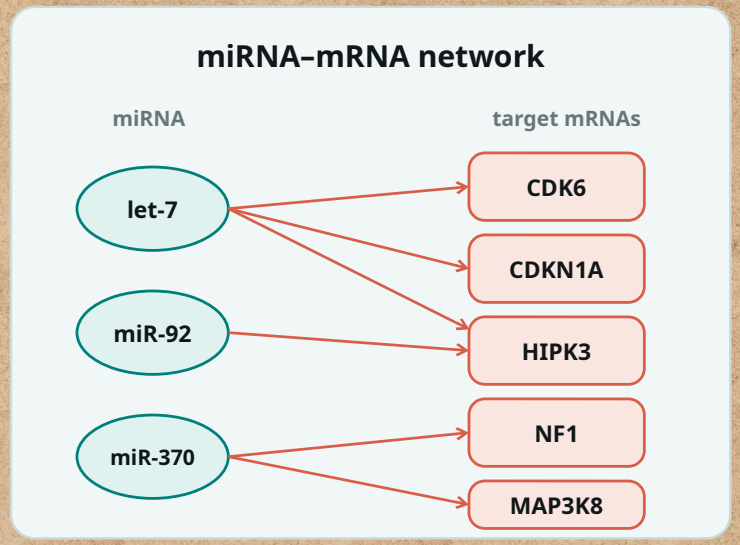
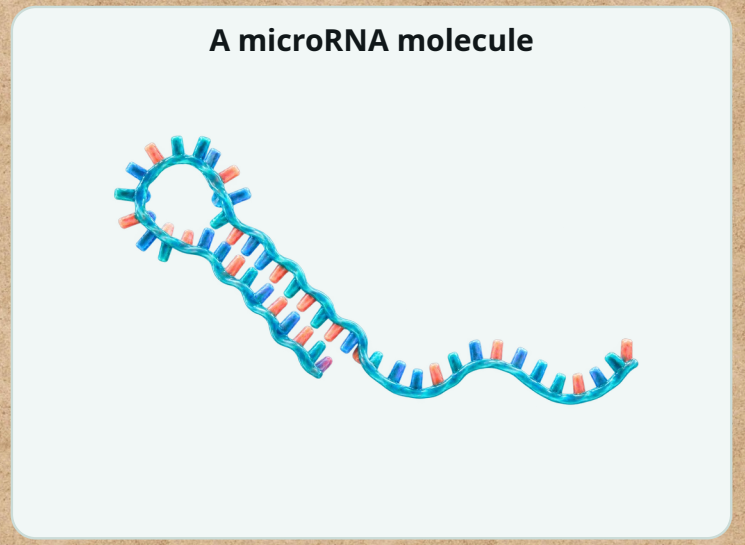
The sound of silencing

Background: microRNA (miRNA) silences gene expression by binding to messenger RNA (mRNA)

Questions: 1.) Do microRNAs regulate genes differently in humans and chimps as the brain develops?
2.) Can that help explain what makes the human brain special?







METHODS

- Time series analysis of weighted regulatory networks
- Functional analysis

DATA

Publicly available miRNA and mRNA expression data from study
Prefrontal cortex (PFC) expression data from humans, chimpanzees and macaques

FIELDS

- Computational neuroscience and genomics
- General network analysis

Human–chimpanzee differences in early development

A) DIFFERENTIAL EXPRESSION

EARLY DEVELOPMENT

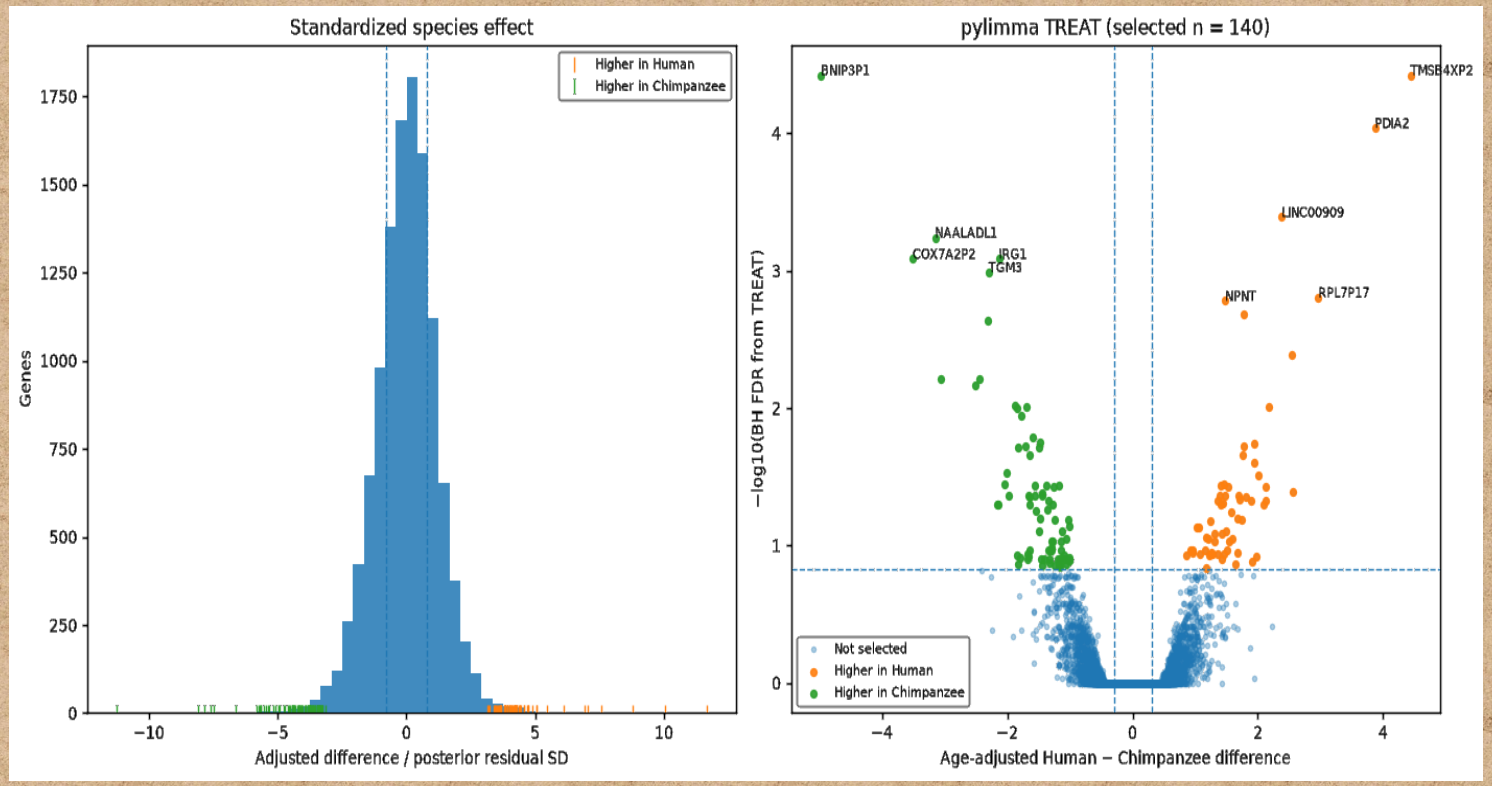
Nine profiles under 100 days:
5 human and 4 chimpanzee.

TOOL

pylimma: Linear Models for Microarray Data.

RESULT

140 of 11,667 genes significant
62 higher in humans; 78 in chimps.



Standardized species effect and TREAT FDR

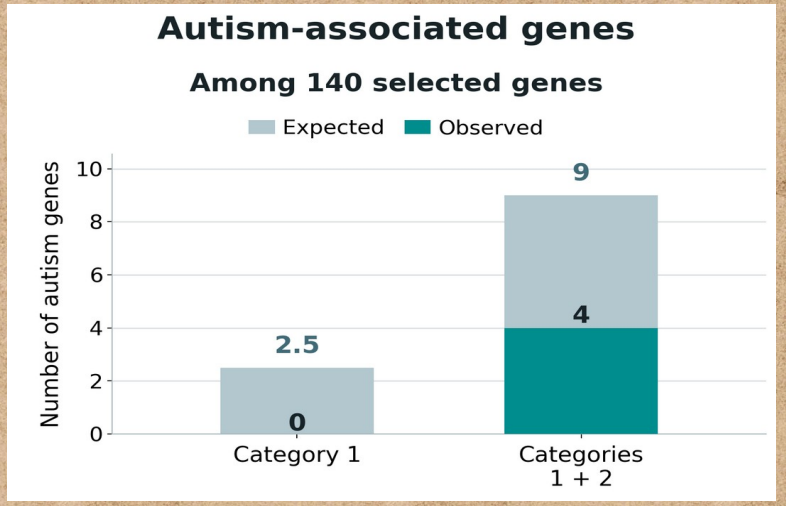
Functional analysis in early development

B) AUTISM-GENE ENRICHMENT

Are autism genes from SFARI database enriched among the 140 early-expression candidates?

FISHER EXACT TEST

No overrepresentation (p = 1.000, one-sided), or underrepresentation (p > 0.1, two-sided)



C) GO BIOLOGICAL PROCESS ANALYSIS

Ranked differential-expression statistics with GOfuncR; FWER ≤ 0.05 (10,000 random sets).

RESULTS AFTER HIERARCHY REFINEMENT

10 human-higher GO categories: transcription, RNA splicing, DNA processes.

30 chimp-higher GO categories: synaptic function, ion transport and mitochondrial electron transport.



Life-history network analysis

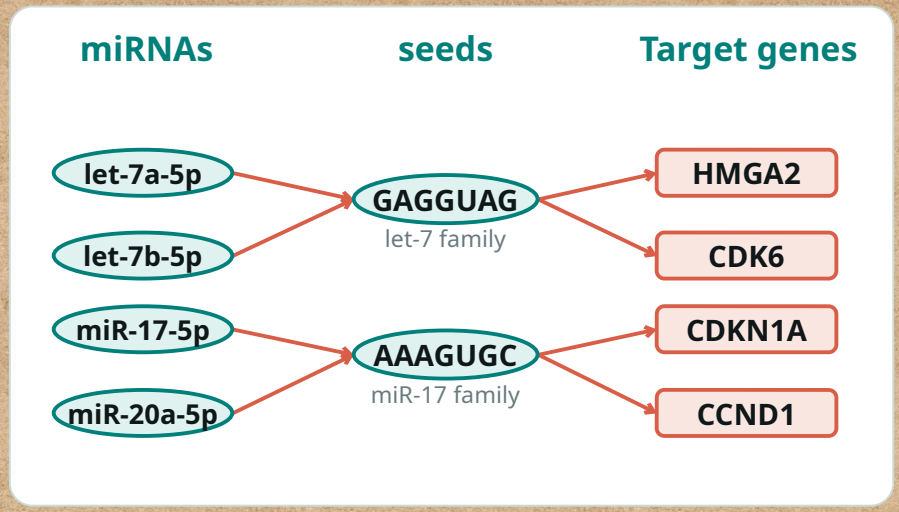
D) LIFE-HISTORY NETWORK ANALYSIS

AGE-MATCHED miRNA-mRNA PROFILES

24 profiles:
7 human, 10 chimpanzee,7 macaque.

NETWORK SIZE

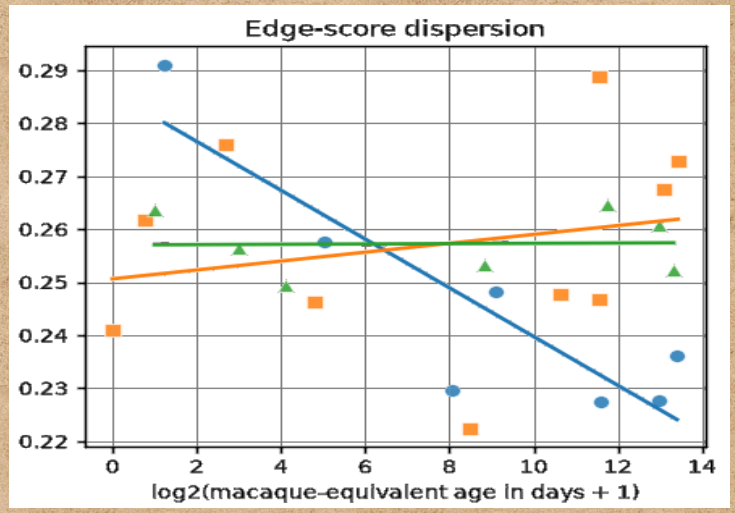
About 200 miRNAs resulted in 150 seeds.
About 7,000 genes and 24,000
seed-gene edges.



MAIN RESULT

Two of six network features, were significant (FDR < 0.05).
Both decline more steeply with age in humans
than in chimps.

This means, with age, the regulatory activity became more evenly distributed across target genes in humans.



Summary and contribution

E) MAIN FINDINGS AND DISTINCTIVE APPROACH

1.) EARLY HUMAN–CHIMP DIFFERENCES

140 genes (62 human-higher; 78 chimp-higher).

AUTISM-GENES

No enrichment of SFARI Category 1 or 1 + 2 genes

GENE ONTOLOGY

10 human-higher categories:
transcription, RNA splicing, DNA processes.

30 chimp-higher categories:
synaptic function, ion transport and mitochondrial electron transport.



2.) NETWORK LIFE-HISTORY ANALYSIS

With age, regulatory activity became more evenly distributed across target genes in the human samples.

SCOPE OF NOVELTY

Instead of analyzing network changes, we kept the network fixed and analyzed global network parameters using regression.